Balancing Performance and Robustness in Automatic Domain Randomization

Michele Carena
Politecnico di Torino
Torino, Italy
s349483@studenti.polito.it

Davide Carletto
Politecnico di Torino
Torino, Italy
s339425@studenti.polito.it

***Abstract*—This project investigates the efficacy of Automatic Domain Randomization (ADR) as a robust adaptive strategy for sim-to-sim transfer in robotic locomotion tasks. While Uniform Domain Randomization (UDR) has shown promise in previous experiments on MuJoCo environments, it relies heavily on manual tuning and fixed parameter distributions, often resulting in suboptimal policies or convergence failures. To address these limitations, we implement ADR, an adaptive algorithm that dynamically calibrates the difficulty of the training distribution by expanding or contracting randomization ranges based on agent performance. We evaluate this approach on two distinct environments, the bipedal Hopper and the quadrupedal Ant, to analyze how ADR adapts to different dynamic challenges. The study aims to validate ADR's ability to eliminate manual tuning, analyze the emergent curriculum dynamics, and assess the trade-off between generalization capabilities and performance efficiency across diverse target configurations.**

***Index Terms*—Deep Reinforcement Learning, Automatic Domain Randomization, Sim-to-Real Transfer**

I. Introduction

Deep Reinforcement Learning (DRL) has demonstrated remarkable capabilities in solving complex robotic locomotion tasks [1], [2]. However, policies trained in pristine simulation environments often fail catastrophically when deployed in target environments with different physical properties, a phenomenon known as the *sim-to-real gap* [3]. To bridge this gap, **Domain Randomization (DR)** has emerged as a standard technique [4], introducing variability in physical parameters (e.g., friction, mass) during training to force the agent to learn robust features.

While **Uniform Domain Randomization (UDR)** is effective [5], it relies heavily on the manual design of randomization ranges. This approach presents a fundamental dilemma: narrow ranges may fail to encompass the target dynamics (overfitting), while excessively wide ranges can make the task unsolvable from the start, preventing the policy from converging. Consequently, finding the best zone for UDR often requires extensive, trial-and-error hyperparameter tuning [6].

To address these limitations, we investigate **Automatic Domain Randomization (ADR)** [7], an adaptive algorithm that automates the curriculum generation. ADR dynamically adjusts the difficulty of the training distribution based on the agent's performance, starting from a single environment and expanding the parameter ranges only when the policy demonstrates proficiency. This curriculum-based approach aims to stabilize training and eliminate the need for manual range tuning.

In this work, we present a comprehensive analysis of ADR applied to robotic locomotion, specifically focusing on the trade-off between randomization aggressiveness and policy generalization. We evaluate our approach on two distinct experimental setups using the **MuJoCo** physics engine [8] via the Gymnasium interface [9]:

1) **Environmental Robustness (Hopper):** We test the agent's ability to adapt to hostile external dynamics, including extreme gravity changes, low-friction surfaces, and small external perturbations.
2) **Morphological Asymmetry (Ant):** We extend the evaluation to internal structural variations by introducing link-level mass perturbations and heavy payload conditions, testing the agent's robustness to severe morphology changes without prior exposure to the specific target configuration.

This work aims to effectively address two aspects of ADR. First, we explore its potential as an automated alternative to UDR, demonstrating its ability to stabilize training in high-dimensional perturbation spaces without requiring pre-defined fixed ranges. Second, through an ablation study on randomization intensity (Easy vs. Medium vs. Hard), we highlight a critical *performance-robustness trade-off*: extending randomization ranges indefinitely produces highly conservative policies that survive worst-case scenarios but suffer reduced efficiency in nominal conditions. Conversely, restricted curricula produce efficient specialists that remain fragile to more hostile environments. These findings suggest that while ADR eliminates manual tuning of fixed ranges, the selection of the target operational envelope remains a crucial design choice for safe deployment.

II. Related Work

A. Domain Randomization in Robotics

Domain Randomization (DR) was originally proposed to bridge the reality gap by training policies over a fixed distribution of physical parameters, ensuring that the real world appears as just another sample from the training set [4]. While effective for locomotion [5] and manipulation, standard Uniform DR (UDR) relies on the manual selection of

randomization bounds. This process is sample-inefficient and prone to human bias: overly wide ranges prevent convergence, while narrow ranges fail to encompass target dynamics [6].

B. Adaptive and Curriculum Learning

To automate range selection, recent works have proposed adaptive strategies. Chebotar et al. [10] utilized real-world data to optimize simulator parameters, a luxury not always available. Alternatively, *Automatic Domain Randomization* (ADR) [7] generates a curriculum by gradually expanding randomization ranges based solely on simulation performance. While Akkaya et al. demonstrated this on dexterous manipulation, our work investigates the application of ADR to locomotion, specifically analyzing the *performance-robustness trade-off* often overlooked in pursuit of pure survival. Recent approaches like DORAEMON [11] propose maximizing entropy to explore diverse dynamics without fixed boundaries.

III. Methodology

While the Introduction outlined the conceptual advantages of ADR, this section details our specific implementation and the design choices made to adapt the algorithm to the Hopper and Ant environments. We utilized the ADR framework [7] integrating it with the **Proximal Policy Optimization (PPO)** algorithm [12].

A. Adaptive Curriculum Mechanism

Unlike Uniform Domain Randomization, where parameter ranges are fixed beforehand, our implementation treats the environment configuration as a **dynamic hyper-rectangle**. For each physical parameter λ_i (e.g., friction, mass), the algorithm maintains a lower bound $\phi_{L,i}$ and an upper bound $\phi_{H,i}$.

The core mechanism is the automatic adjustment of these bounds based on the agent's performance buffer. At fixed intervals, the algorithm evaluates the policy's average reward $\bar{R}$. The update rule for a boundary ϕ (applies symmetrically to both ϕ_L and ϕ_H) follows a threshold logic to continuously prompt the agent:

$$\phi^{t+1} = \begin{cases} \phi^t + \delta & \text{if } \bar{R} \geq T_{perf} \\ \phi^t - \delta & \text{if } \bar{R} < T_{perf} \end{cases} \tag{1}$$

where δ is the step size (expansion rate) and T_{perf} is the target performance threshold, defined as a percentage of the optimal baseline return. This binary logic creates a "breathing" curriculum: the ranges expand when the agent masters the current difficulty, and immediately contract if performance drops, ensuring the training distribution remains exactly at the edge of the agent's capabilities without stagnation.

B. Sampling Strategies: Boundary vs. Vanilla

A critical design choice in ADR is how to sample parameters within the current dynamic ranges $[\phi_L, \phi_H]$ during training episodes. In this work, we employed two distinct strategies tailored to the specific complexity of each robot:

- **Boundary Sampling (Hopper):** For the Hopper model we utilized the Boundary Sampling technique. This method prioritizes sampling parameters from the extreme edges of the current range (the lower bound ϕ_L or upper bound ϕ_H) with a probability p_b. This focuses the learning signal on the "frontier" of feasibility, accelerating the expansion of the validity domain.
- **Vanilla Sampling (Ant):** For the more complex Ant model, we opted for a *Vanilla* approach. In this configuration, parameters are always sampled uniformly from the entire current volume ($p_b = 0$). Given the higher dimensionality and intrinsic instability of the Ant's dynamics under morphology perturbations, uniform sampling provides a smoother curriculum, preventing the policy from collapsing due to excessive stress from edge-case combinations.

C. Curriculum Aggressiveness

To investigate the trade-off between robustness and performance, we did not use a single set of hyperparameters. Instead, we defined three distinct ADR configurations characterized by increasing "aggressiveness":

- **Easy ADR:** Uses a small step size δ and conservative thresholds. This mimics a safe, slow curriculum intended to preserve high performance on nominal tasks.
- **Medium ADR:** Balances expansion speed and training stability.
- **Hard ADR:** Uses a larger δ (e.g., 0.05 vs 0.02) and wider maximum limits. This forces the agent to adapt to rapid changes in dynamics, prioritizing worst-case survival over efficiency.

The specific parameter sets, bounds, and thresholds for both environments are detailed in the Experimental Setup section.

IV. Experimental Setup

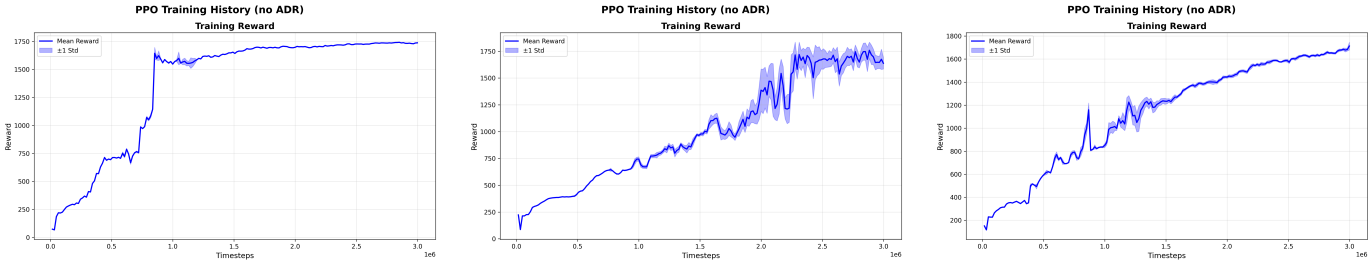
To validate the proposed methodology, we designed a rigorous evaluation protocol involving two distinct locomotion tasks. For both environments, we first established a performance ceiling by training standard PPO agents directly on the target configurations **(Upper Bound)**. These values were used to calibrate the ADR performance thresholds (T_{perf}), ensuring the curriculum expanded only when the agent achieved a significant fraction of the optimal possible return.

A. Environmental Robustness (Hopper)

The Hopper experiments focused on the agent's ability to withstand external dynamic variations using **Boundary Sampling**.

Target Configurations: We defined a nominal Source domain and three Target domains of increasing hostility to test generalization:

- **Target-Easy:** Introduces a minor morphological discrepancy by reducing the torso mass by 1.0 kg compared to the source.



(a) Target-Easy (b) Target-Medium (c) Target-Hard

Fig. 1: Training progression of Upper Bound baselines. **Target-Easy** shows rapid convergence with minimal variance. **Target-Medium** exhibits noticeable instability (wider standard deviation bands) due to friction ambiguity, whereas **Target-Hard** displays a slower but surprisingly more stable convergence, suggesting the adoption of a consistent, conservative gait.

- **Target-Medium:** Adds moderate randomization, varying link masses by $\pm 20\%$ and reducing ground friction to $0.7\times$.
- **Target-Hard:** Represents an extreme environment. It involves mass variations of $\pm 50\%$, low friction ($0.5\times$), high gravity (-11 m/s^2), and random pushes (Force 10.0N).

ADR Training configurations: We trained three ADR agents, scaling the training duration and curriculum aggressiveness according to the target difficulty. We used the following configurations derived from our ablation study:

- **Easy ADR:** Trained for 5M timesteps with conservative parameters ($\delta = 0.03$) and a low boundary sampling probability ($p_b = 0.1$).
- **Medium ADR:** Trained for 7M timesteps with balanced settings ($\delta = 0.03, p_b = 0.3$).
- **Hard ADR:** Trained for 8M timesteps with aggressive expansion ($\delta = 0.07$) and frequent boundary sampling ($p_b = 0.5$).

The thresholds T_{perf} were set to 70%, 60%, and 60% of the respective Upper Bound baselines.

B. Morphological Robustness (Ant)

The Ant experiments extended the evaluation to robustness against internal mass perturbations using **Vanilla Sampling** (uniform distribution) to maintain stability.

Target Design (Hell Mode): We defined a single, highly challenging evaluation domain named *Hell Mode*. This environment introduces severe coupled perturbations: increased torso mass ($2.5\times$), unbalanced internal link masses applied to a subset of limb segments (some links increased by +50% and others reduced by -50%), reduced friction ($0.5\times$), strong gravity (-12 m/s^2), and random external pushes. The baseline Upper Bound return on this domain was established at **2370.0**.

ADR Variants: We compared three ADR variants to isolate the effects of different randomization sets:

- **Ant-Minimal (Easy):** Randomizes only global parameters (friction, damping, gravity). This tests whether environmental noise alone aids transfer under mass imbalance.
- **Ant-Asymmetric (Medium):** Extends the minimal configuration by including a subset of internal link mass multipliers, targeting robustness to unbalanced internal dynamics.
- **Ant-Heavy (Hard):** Randomizes the complete set of mass-related parameters defined in our model (torso and selected limb links), together with friction, damping and gravity, exposing the policy to high-dimensional internal perturbations.

Transfer Protocol: Given the high dimensionality of the Ant dynamics, determining an optimal fixed performance threshold (T_{perf}) *a priori* presents a challenge. Setting this value incorrectly could lead to premature curriculum expansion while the agent is still mastering basic coordination. To mitigate this and strictly decouple the acquisition of locomotion skills from the robustification phase, we adopted a **two-stage pipeline**: agents were pre-trained on the nominal source domain to establish a stable gait baseline. Subsequently, ADR was activated for an additional 2M timesteps, with curriculum ranges initialized at nominal values and allowed to expand based on a calibrated threshold T_{perf} (45% of the Upper Bound).

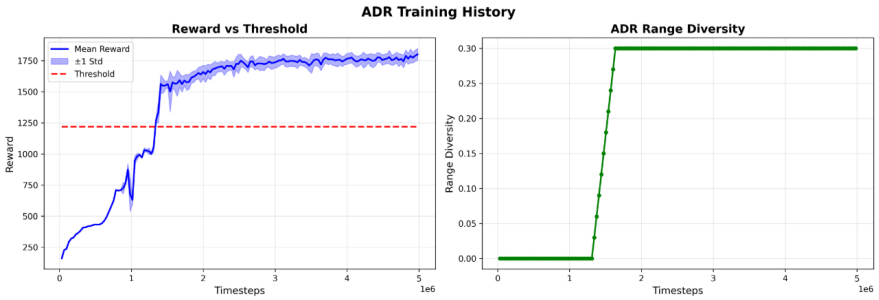
V. Results

A. Environmental Robustness (Hopper)

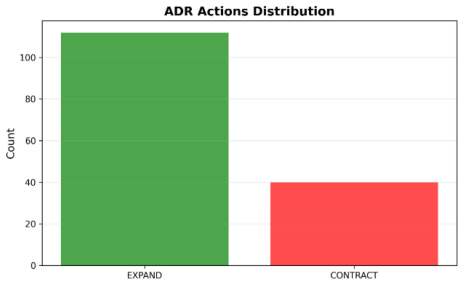
To analyze the generalization capabilities of the learned policies, we conducted a comprehensive **cross-evaluation**. We tested each ADR agent (Easy, Medium, Hard) against all three Target environments. Additionally, we compared these results against the **Upper Bound** baselines (standard PPO trained directly on each target). The learning curves establishing these baselines are depicted in Fig. 1.

The detailed evaluation results are summarized in Table I and visualized in Fig. 4.

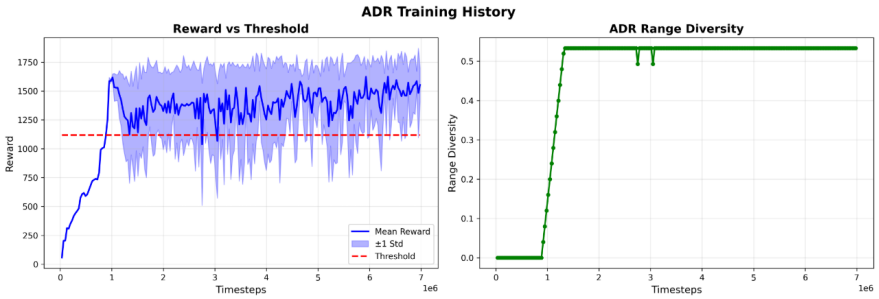
ADR Training Dynamics: To understand the emergence of these distinct behaviors, we analyze the training progression in Fig. 2. The *ADR Easy* variant (Top Row) exhibits monotonic range expansion with negligible reward variance, indicating the curriculum remained within the agent's "comfort zone." *ADR Medium* variant (Middle Row) displays increased variance due to reduced friction, yet the agent successfully



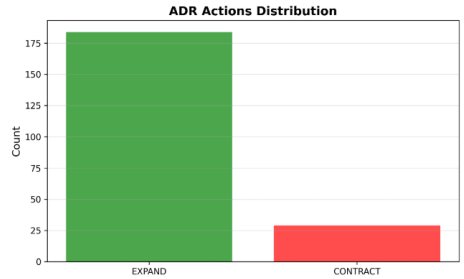
(a) Hopper-ADR Easy: Training History



(b) Hopper-ADR Easy: Action Distribution

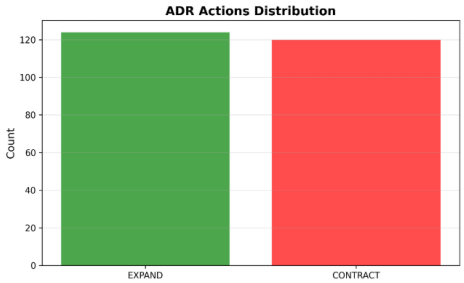


(c) Hopper-ADR Medium: Training History



(d) Hopper-ADR Medium: Action Distribution

(e) Hopper-ADR Hard: Training History



(f) Hopper-ADR Hard: Action Distribution

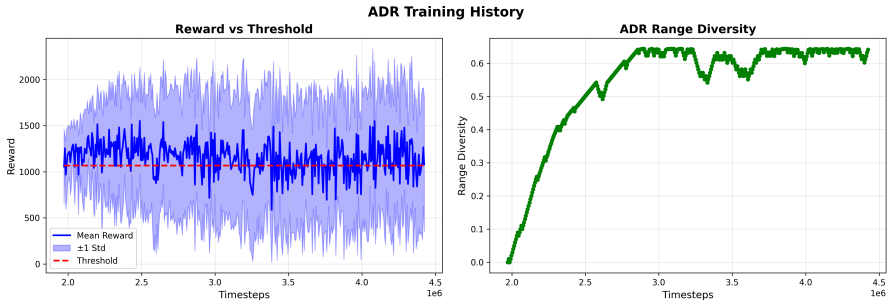
Fig. 2: ADR training dynamics for Hopper variants. **Top Row (Easy):** Shows monotonic range expansion and high stability (narrow reward variance), indicating the curriculum remained within the agent's "comfort zone". **Middle Row (Medium):** Exhibits increased variance due to reduced friction, but the agent successfully expands ranges to saturation. **Bottom Row (Hard):** Illustrates the "breathing" nature of the curriculum. The jagged range evolution (Left) and the balanced Action Distribution (Right, nearly 50/50 split) confirm that ADR actively regulated difficulty—alternating expansion and contraction—to prevent collapse in this hostile domain.

drives the curriculum to saturation limits. In contrast, the *ADR Hard* variant (Bottom Row) displays a "jagged" range evolution and a balanced action distribution (nearly 50% contractions). This confirms that the curriculum actively regulated difficulty—alternating expansion and contraction—to prevent collapse. This struggle at the "frontier of feasibility" forced the agent to adopt the conservative gait responsible for its survival in extreme conditions.

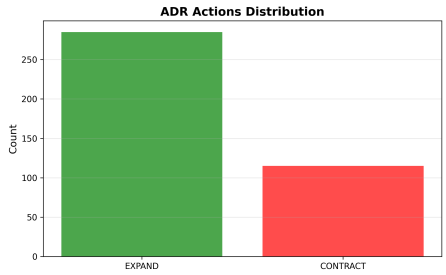
TABLE I: Hopper Cross-Evaluation (Mean Reward $\pm$ Std)

Training Setup	Target Easy	Target Medium	Target Hard
Upper Bound (PPO)	*1740.9 $\pm$ 2.1*	*1645.3 $\pm$ 59.3*	*1710.8 $\pm$ 40.6*
ADR Easy	**1780.3 $\pm$ 9.4**	567.5 $\pm$ 52.9	383.3 $\pm$ 18.7
ADR Medium	1582.0 $\pm$ 182.4	**1500.4 $\pm$ 16.1**	727.2 $\pm$ 140.9
ADR Hard	1430.6 $\pm$ 36.6	1476.2 $\pm$ 33.1	**1353.9 $\pm$ 96.7**

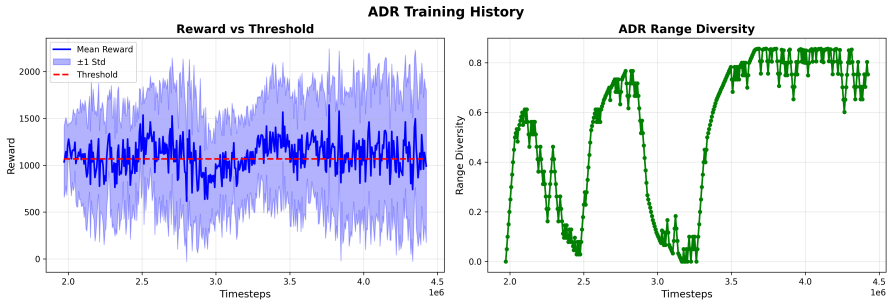
The Performance-Robustness Trade-off: The evaluation on **Target-Easy** reveals a crucial insight. The **ADR Easy** agent slightly exceeds the Upper Bound (1780 vs 1740), having effectively mastered the nominal dynamics. However, as the curriculum aggressiveness increases, performance on this simple task degrades: **ADR Medium** drops to 1582, and **ADR Hard** further to 1430. This gradient quantitatively demonstrates the *cost of robustness*: the conservative gait required



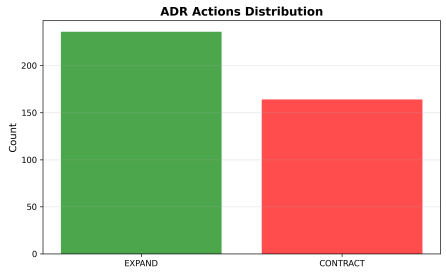
(a) Ant-Minimal: Training History



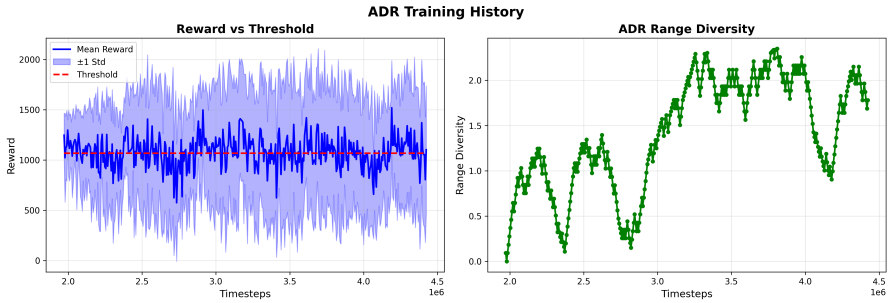
(b) Ant-Minimal: Action Distribution



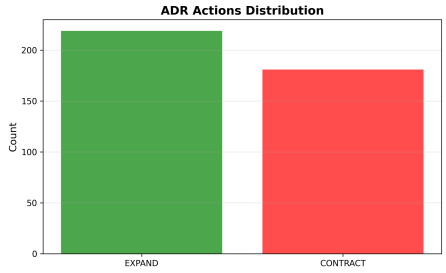
(c) Ant-Asymmetric: Training History



(d) Ant-Asymmetric: Action Distribution



(e) Ant-Heavy: Training History



(f) Ant-Heavy: Action Distribution

Fig. 3: Detailed ADR dynamics for each Ant variant. Each row presents the training evolution (Left) and the corresponding curriculum action distribution (Right). **Top Row (Minimal):** Shows high variance and rapid expansion but lacks stability. **Middle Row (Asymmetric):** Shows efficient learning but catastrophic fragility (low min reward). **Bottom Row (Heavy):** Demonstrates the "Workhorse" behavior: frequent contractions (orange bars) ensure the agent remains within feasible limits while mastering the heavy payload.

to survive the Hard curriculum is inherently suboptimal for benign environments.

Generalization Capabilities: The limitations of the specialist become evident on harsher domains:

- **Target-Medium:** The ADR Easy policy suffers a catastrophic collapse ($\sim$ 567), proving that policies overfitted to high friction cannot generalize even to moderate shifts. The **ADR Hard** model, however, shows excellent *backward generalization* (1476), nearly matching the specialist.
- **Target-Hard:** On this extreme domain, only the **ADR Hard** agent remains viable (1353). The gap compared to the Upper Bound (1710) reflects the difficulty of learning a single policy that covers the entire spectrum of physics.

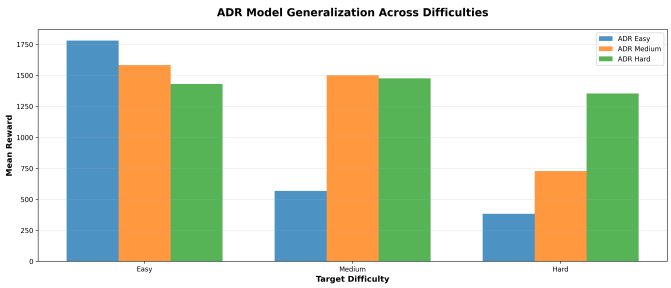


Fig. 4: ADR Performance comparison on Hopper Targets

B. Morphological Robustness (Ant)

For the Ant, we evaluated the three variants on the single **"Hell Mode"** target. Table II presents the results, while Fig. 3 details the training dynamics. Unlike Hopper, Mean Reward alone is insufficient here; we introduce *Minimum Reward* to assess reliability against catastrophic failures.

TABLE II: Ant Performance on "Hell Mode" (100 Episodes)

Model	Mean Reward	Min Reward	Success Rate
Upper Bound	*2370.0*	-	*100%*
Ant-Minimal (Easy)	1150 ± 458	210	100%
Ant-Asymmetric (Med)	$\mathbf{1441} \pm 120$	**16**	85%
Ant-Heavy (Hard)	1233 ± 180	**319**	**100%**

Behavioral Analysis: The results highlight a divergence in strategies, which can be directly traced back to the ADR dynamics shown in Fig. 3:

- **Ant-Asymmetric (The "Glass Cannon"):** This model achieves the highest Mean Reward (1441), indicating an efficient sprinting gait. However, since torso mass was excluded from training, the unseen payload destabilizes it. This fragility is evident in the low Min Reward (16) and the binary nature of its failure cases.
- **Ant-Minimal (The "Survivor"):** Despite never experiencing mass perturbations, this agent attains 100% success. As seen in Fig. 3 (Top Row), its training was characterized by high variance. This reflects *accidental robustness*: the friction randomization forced a cautious policy that avoids unstable motions, but the high standard deviation (± 458) suggests it is "fighting" the dynamics rather than mastering them.
- **Ant-Heavy (The "Workhorse"):** This agent converges to a conservative but stable strategy with the highest Minimum Reward (319). The training history in Fig. 3 (Bottom Row) explains why: the frequent curriculum contractions (orange bars) indicate that the ADR mechanism successfully identified the stability boundaries of the heavy payload, constraining the learning process to feasible regions to ensure safety.

VI. Conclusions

In this work, we presented a comprehensive evaluation of **Automatic Domain Randomization (ADR)** as a mechanism to bridge the sim-to-sim gap in robotic locomotion. By dynamically coupling the training distribution to the agent's performance, our implementation successfully eliminated the manual tuning typical of Uniform Domain Randomization, preventing convergence failures in high-dimensional perturbation spaces.

Our experimental analysis yielded two fundamental insights. First, we identified a distinct **Performance-Robustness Trade-off**. The cross-evaluation on the Hopper environment demonstrated that unconstrained range expansion is not strictly optimal; while "Hard" curricula produce agents capable of surviving extreme worst-case scenarios (ice, high gravity), they induce conservative policies that underperform in benign conditions compared to specialized agents. This suggests that robustness incurs a "performance tax" on efficiency.

Second, the experiments on the Ant validated the **versatility** of ADR across different types of generalization. We showed that the method effectively scales from environmental parameter mismatches to severe morphological variations. The contrast between the "Glass Cannon" behavior of the Asymmetric agent and the stability of the Heavy agent highlights that robustness against internal dynamics (e.g., payloads, limb masses) requires explicit inclusion in the randomization set; environmental noise alone is insufficient to guarantee safety against structural changes.

We conclude that while ADR automates curriculum generation, defining the *target operational envelope* remains a crucial design choice. Future work should investigate multi-objective curricula or generative approaches like **DORAEMON** [11]. Unlike ADR's bounded hyper-rectangles, such strategies maximize entropy to autonomously discover challenging "unknown unknowns" without pre-specified constraints.

References

[1] V. Mnih, K. Kavukcuoglu, D. Silver *et al.*, "Human-level control through deep reinforcement learning," *nature*, vol. 518, no. 7540, pp. 529–533, 2015.

[2] T. P. Lillicrap, J. J. Hunt, A. Pritzel, N. Heess, T. Erez, Y. Tassa, D. Silver, and D. Wierstra, "Continuous control with deep reinforcement learning," *arXiv preprint arXiv:1509.02971*, 2015.

[3] W. Zhao, J. P. Queralta, and T. Westerlund, "Sim-to-real transfer in deep reinforcement learning for robotics: a survey," *2020 IEEE Symposium Series on Computational Intelligence (SSCI)*, pp. 737–744, 2020.

[4] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain randomization for transferring deep neural networks from simulation to the real world," *2017 IEEE/RSJ international conference on intelligent robots and systems (IROS)*, pp. 23–30, 2017.

[5] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, "Sim-to-real transfer of robotic control with dynamics randomization," *2018 IEEE international conference on robotics and automation (ICRA)*, pp. 3803–3810, 2018.

[6] Q. Vuong, S. Vikram, H. Su, S. Gao, and H. I. Christensen, "How to pick the domain randomization parameters for sim-to-real transfer of reinforcement learning policies?" *arXiv preprint arXiv:1903.11774*, 2019.

[7] I. Akkaya, M. Andrychowicz, M. Chociej, M. Chrzanowski, D. Kumar, B. Dormandez *et al.*, "Solving rubik's cube with a robot hand," *arXiv preprint arXiv:1910.07113*, 2019.

[8] E. Todorov, T. Erez, and Y. Tassa, "Mujoco: A physics engine for model-based control," in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2012, pp. 5026–5033.

[9] G. Brockman, V. Cheung, L. Pettersson, J. Schneider, J. Schulman, J. Tang, and W. Zaremba, "Openai gym," *arXiv preprint arXiv:1606.01540*, 2016.

[10] Y. Chebotar, A. Handa, V. Makoviychuk, M. Macklin, J. Issac, N. Ratliff, and D. Fox, "Closing the sim-to-real loop: Adapting simulation randomization with real world experience," in *2019 International Conference on Robotics and Automation (ICRA)*. IEEE, 2019, pp. 3273–3279.

[11] G. Tiboni, P. Klink, J. Peters, T. Tommasi, C. D'Eramo, and G. Chalvatzaki, "DORAEMON: Domain randomization via entropy maximization," in *International Conference on Learning Representations (ICLR)*, 2024.

[12] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.